Topic quiz: Momentum and impulse

IBDP Physics SL · Grade 11 · A.2 Momentum · 10 multiple-choice questions

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Name: Class: Date:

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use g = 9.81 m s^{-2} and show any working next to the question.
Check yourself online: the same quiz with instant feedback and explanations — https://school.taneski.com/physics/dp/quiz-momentum/

1	2	3	4	5	6	7	8	9	10

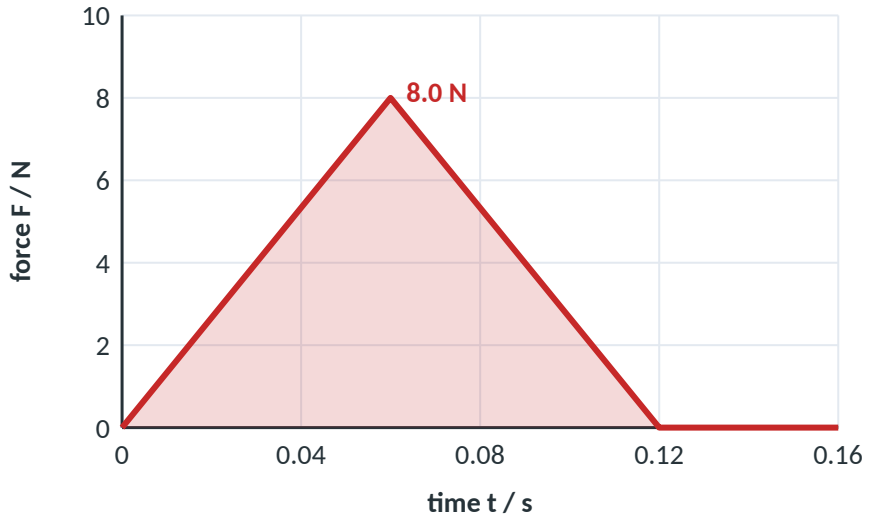
Question 1

A 0.500 kg cart moving at +0.62 m s^{-1} hits a force sensor and rebounds at 0.50 m s^{-1}. What is the impulse on the cart?

(A) +0.31 N s
(B) −0.06 N s
(C) −0.56 N s
(D) +0.56 N s

Question 2

The graph shows the force on a ball while it is hit. What is the impulse on the ball?



(A) 0.48 N s
(B) 0.96 N s
(C) 8.0 N s
(D) 67 N s

Question 3

An egg thrown into a loose bedsheet does not break, but the same egg thrown at a wall does. Why?

(A) The sheet makes the contact time longer, so the average force is smaller
(B) The sheet makes the egg's change in momentum smaller
(C) The sheet makes the impulse on the egg smaller
(D) The sheet absorbs the egg's momentum, so momentum is not conserved

Question 4

A 0.058 kg tennis ball arrives at a racket at 20 m s^{-1} and leaves in the opposite direction at 30 m s^{-1}. The contact time is 5.0 ms. What is the average force of the racket on the ball?

(A) 0.58 N
(B) 2.9 N
(C) 116 N
(D) 580 N

Question 5

A 0.25 kg cart moving at 0.60 m s^{-1} hits a stationary 0.50 kg cart. They stick together. What is their speed after the collision?

(A) 0.15 m s^{-1}
(B) 0.20 m s^{-1}
(C) 0.30 m s^{-1}
(D) 0.60 m s^{-1}

Question 6

In the collision of the last question, what fraction of the kinetic energy is transferred to other forms?

(A) 0 %, because momentum is conserved
(B) 33 %
(C) 67 %
(D) 100 %

Question 7

A 200 kg cannon at rest fires a 5.0 kg shell forwards at 120 m s^{-1}. What is the recoil velocity of the cannon?

(A) 3.0 m s^{-1} forwards
(B) 3.0 m s^{-1} backwards
(C) 2.9 m s^{-1} backwards
(D) 120 m s^{-1} backwards

Question 8

A heavy truck collides head-on with a light car. How does the force of the truck on the car compare with the force of the car on the truck?

(A) The force on the car is larger
(B) The force on the truck is larger
(C) It depends on which one is moving faster
(D) They are equal in size and opposite in direction

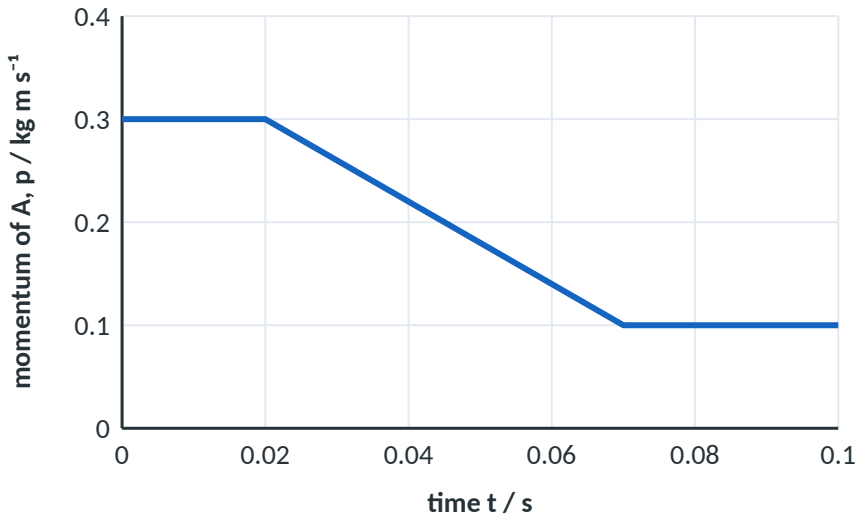
Question 9

Which quantities are conserved in a perfectly elastic collision between two carts?

(A) Total momentum only
(B) Total kinetic energy only
(C) Both total momentum and total kinetic energy
(D) Neither — some energy always becomes heat

Question 10

The graph shows the momentum of cart A during a collision. What is the average force on cart A during the collision?



(A) +4.0 N

(B) −4.0 N

(C) −0.20 N

(D) −0.010 N

End of quiz · Score: _____ / 10